



National University of Modern Languages, Islamabad

LAB 5

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Subject:	MAD
Due Date:	10th November 2025
Semester:	8th
Course:	BSCS, Morning

Faculty of Engineering and Computer Science

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1. Lab Task for Attendance App

Design a Flutter Attendance app that demonstrates state management using the Provider package with ChangeNotifier and Consumer. Create an AttendanceModel that maintains a list of students (each having {id, name, isPresent}) and provides methods to toggle a student's attendance status and mark all present or all absent. Build a UI displaying a scrollable list of students with checkboxes to mark attendance, along with a counter showing the total number of present students. Ensure that only the widgets dependent on changing values (such as individual list items and the counter) rebuild when the state updates, while the rest of the widget tree (e.g., app bar, static layout) remains unaffected. The app should clearly demonstrate efficient state management with ChangeNotifierProvider, Consumer, and selective rebuilding.

1.1 Description

This Flutter Attendance App is built using the **MVC architecture** with **Provider state management**.

It efficiently tracks student attendance through a clean, responsive interface.

- **Model (Student.dart)** – Defines the student data structure (id, name, isPresent).
- **Controller (attendance_controller.dart)** – Handles user interactions and updates the provider's state (e.g., toggling attendance).
- **Provider (attendance_provider.dart)** – Manages and notifies UI about state changes such as toggling individual attendance, marking all absent, and counting total presents.
- **View (attendance_screen.dart)** – Displays a scrollable list of students with animated attendance switches, a styled counter showing total present students, and a responsive button to mark all absent.
- **Main (main.dart)** – Initializes the ChangeNotifierProvider and sets up the app theme and home screen.

The app demonstrates **efficient state management**, ensuring only the widgets depending on changing data (like switches and counters) rebuild, while static UI elements remain unaffected.

1.2 Code

1.2.1 Model: Student.dart

```
// models/student.dart
```

```
class Student {  
  
  final int id;  
  
  final String name;  
  
  bool isPresent;  
  
  Student({  
    required this.id,  
    required this.name,  
    this.isPresent = false,  
  });  
}
```

1.2.2 Controller: attendance_controller.dart

```
// controllers/attendance_controller.dart
```

```
import 'package:flutter/material.dart';  
import 'package:provider/provider.dart';  
import '../providers/attendance_provider.dart';  
  
class AttendanceController {  
  
  void markAttendance(BuildContext context, int id) {  
    Provider.of<AttendanceProvider>(context, listen: false).toggleAttendance(id);  
  }  
  
  int getTotalPresent(BuildContext context) {  
    return Provider.of<AttendanceProvider>(context, listen: false).totalPresent;  
  }  
}
```

```
}
```

1.2.3 Provider: attendance_provider.dart

```
// providers/attendance_provider.dart

import 'package:flutter/foundation.dart';
import '../models/student.dart';

class AttendanceProvider extends ChangeNotifier {
  final List<Student> _students = [
    Student(id: 1, name: 'Ali '),
    Student(id: 2, name: 'Ahmed'),
    Student(id: 3, name: 'Hamza'),
    Student(id: 4, name: 'Haider'),
    Student(id: 5, name: 'Umer'),
  ];

  List<Student> get students => List.unmodifiable(_students);

  void toggleAttendance(int id) {
    final index = _students.indexWhere((s) => s.id == id);
    if (index != -1) {
      _students[index].isPresent = !_students[index].isPresent;
      notifyListeners();
    }
  }

  void markAllAbsent() {
    for (var student in _students) {
      student.isPresent = false;
    }
  }
}
```

```
        notifyListeners();
    }

    int get totalPresent => _students.where((s) => s.isPresent).length;
}
```

1.2.4 View: attendance_screen.dart

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
import '../providers/attendance_provider.dart';
import '../controllers/attendance_controller.dart';

class AttendanceScreen extends StatelessWidget {
    final AttendanceController _controller = AttendanceController();

    AttendanceScreen({super.key});

    @override
    Widget build(BuildContext context) {
        final attendanceProvider = Provider.of<AttendanceProvider>(context);

        return Scaffold(
            backgroundColor: Colors.purple.shade50,
            appBar: AppBar(
                title: const Text(
                    "Attendance List",
                    style: TextStyle(
                        fontWeight: FontWeight.bold,
                        fontSize: 22,
```

```
    ),  
    ),  
    centerTitle: true,  
    elevation: 4,  
    flexibleSpace: Container(  
      decoration: const BoxDecoration(  
        gradient: LinearGradient(  
          colors: [Color(0xFF8E24AA), Color(0xFFBA68C8)],  
          begin: Alignment.topLeft,  
          end: Alignment.bottomRight,  
        ),  
      ),  
    ),  
    shadowColor: Colors.purpleAccent.withOpacity(0.3),  
  ),  
  
  body: Column(  
    children: [  
      const SizedBox(height: 8),  
  
      Expanded(  
        child: ListView.builder(  
          padding: const EdgeInsets.symmetric(vertical: 8),  
          itemCount: attendanceProvider.students.length,  
          itemBuilder: (context, index) {  
            final student = attendanceProvider.students[index];  
  
            return AnimatedContainer(  

```

```
duration: const Duration(milliseconds: 300),
margin: const EdgeInsets.symmetric(
  horizontal: 16,
  vertical: 6,
),
decoration: BoxDecoration(
  color: student.isPresent
    ? Colors.purple.shade100
    : Colors.white,
  borderRadius: BorderRadius.circular(14),
  border: Border.all(
    color: student.isPresent
      ? Colors.purple.shade300
      : Colors.purple.shade100,
    width: 1.2,
  ),
  boxShadow: [
    BoxShadow(
      color: Colors.purple.withOpacity(0.08),
      blurRadius: 5,
      offset: const Offset(0, 3),
    ),
  ],
),
child: ListTile(
  contentPadding:
    const EdgeInsets.symmetric(horizontal: 20, vertical: 6),
  title: Text(
```



```

        student.name,
        style: TextStyle(
          fontWeight: FontWeight.w600,
          fontSize: 18,
          color: student.isPresent
            ? Colors.purple.shade800
            : Colors.grey.shade800,
        ),
      ),
      trailing: Switch(
        value: student.isPresent,
        activeColor: Colors.purple,
        activeTrackColor: Colors.purple.shade200,
        inactiveThumbColor: Colors.purple.shade100,
        inactiveTrackColor: Colors.grey.shade300,
        onChanged: (_) {
          _controller.markAttendance(context, student.id);
        },
      ),
    ),
  );
},
),
),
Container(
  margin: const EdgeInsets.all(16),
  padding: const EdgeInsets.symmetric(horizontal: 20, vertical: 16),

```

```
decoration: BoxDecoration(
  color: Colors.white,
  borderRadius: BorderRadius.circular(16),
  boxShadow: [
    BoxShadow(
      color: Colors.purple.withOpacity(0.15),
      blurRadius: 8,
      offset: const Offset(0, 4),
    ),
  ],
),
child: Column(
  crossAxisAlignment: CrossAxisAlignment.stretch,
  children: [

    Row(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [

        const Icon(Icons.people_alt_rounded,
          color: Colors.purple, size: 26),
        const SizedBox(width: 10),
        Flexible(
          child: Text(
            "Total Present: ${attendanceProvider.totalPresent}",
            style: const TextStyle(
              fontSize: 20,
              fontWeight: FontWeight.bold,
              color: Colors.purple,
```

```
letterSpacing: 0.5,
),
overflow: TextOverflow.ellipsis,
),
),
],
),

const SizedBox(height: 12),

SizedBox(
width: double.infinity,
child: ElevatedButton.icon(
onPressed: () {
attendanceProvider.markAllAbsent();
},
icon: const Icon(Icons.cancel, color: Colors.white),
label: const Text(
"Mark All Absent",
style: TextStyle(
fontWeight: FontWeight.bold,
fontSize: 16,
),
),
style: ElevatedButton.styleFrom(
backgroundColor: Colors.redAccent,
foregroundColor: Colors.white,
padding: const EdgeInsets.symmetric(
```

```

        vertical: 14,
      ),
      shape: RoundedRectangleBorder(
        borderRadius: BorderRadius.circular(30),
      ),
      elevation: 5,
    ),
  ),
),
],
),
)

],
),
);
}
}

```

1.2.5 Main: main.dart

```

// main.dart

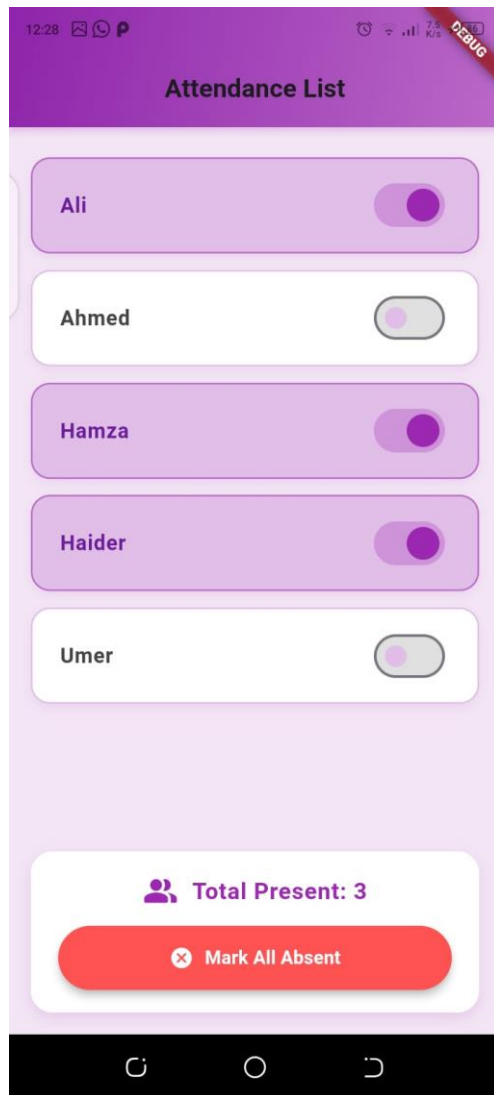
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
import 'providers/attendance_provider.dart';
import 'views/attendance_screen.dart';

void main() {
  runApp(const MyApp());
}

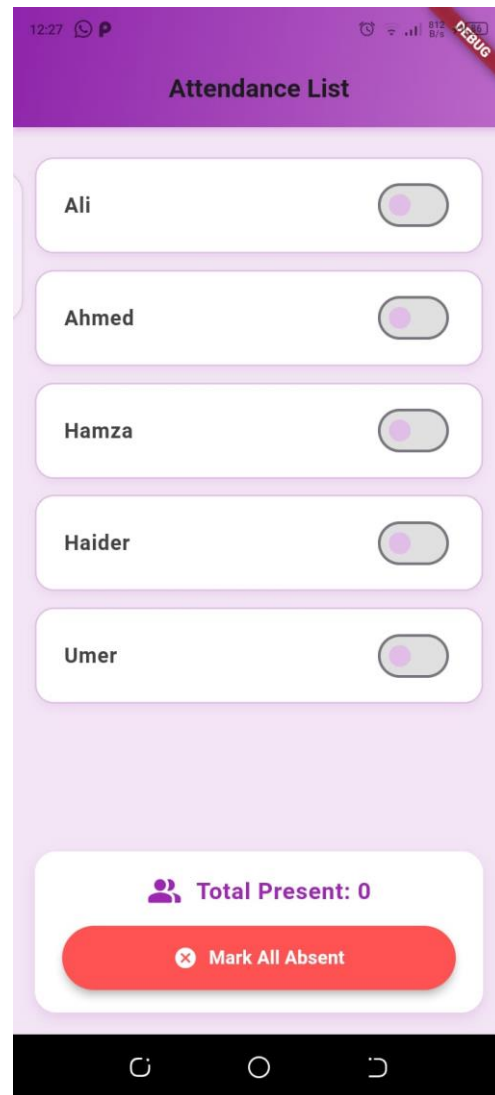
```

```
class MyApp extends StatelessWidget {  
  const MyApp({super.key});  
  @override  
  Widget build(BuildContext context) {  
    return ChangeNotifierProvider(  
      create: (_) => AttendanceProvider(),  
      child: MaterialApp(  
        title: 'Attendance App',  
        theme: ThemeData(  
          primarySwatch: Colors.teal,  
        ),  
        home: AttendanceScreen(),  
      ),  
    );  
  }  
}
```

1.3 Screenshots



Marked two students present



App view after marking all absent

